# Harshita Adlakha

## Making AI safe enough to fly.

ECE undergraduate researching verification of AI in safety-critical aerospace systems.

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### EDUCATION

**Bangalore Institute of Technology**

B.Tech, Electronics & Communication Engineering
VTU · 2024–2028 · GPA 7.9

### PUBLICATIONS

An Architectural Lifecycle Framework for the Verification and Validation of AI-Enabled Aerospace Systems
AIAA SciTech 2027 Forum · accepted

A Sociotechnical Framework for Verifying Non-Deterministic AI in Safety-Critical Aviation Electronics
AIAA SciTech 2027 Forum · accepted

Socio-Technical Complexity Management in Next-Generation Avionics Systems
IAF 77th International Astronautical Congress · Oct 2026

### CERTIFICATIONS

Google Cloud Certified: Generative AI Leader
IBM Qiskit Advocate
Airtable: Agentic Systems Design
IBM SkillsBuild: Software Engineering
Google Cloud Professional Cloud Architect (in progress)

### SKILLS

Python · C/C++ · Verilog · MATLAB · SQL
PyTorch · XGBoost · SHAP · LLMs · RAG
GCP · Vertex AI · Cloud Run · ADK
MBSE · systems architecture · Git

### EXPERIENCE

**Research Associate, Blue Marble Space Young Scientist Program**

Jun–Aug 2026 · under Dr. Jen Blank

AI-assisted design of human habitats on the Moon and Mars: systems-level concepts using in-situ resource utilisation. Invited to present at BlueSciCon Forum 2026.

### PROJECTS

**FDIR Copilot · satellite fault diagnosis agent**

Accel AI Innovate Paris hackathon · Oct 2026 · Python, FastAPI, PostgreSQL

Finds the root cause of small-satellite anomalies from a causal graph and incident memory. 93% root-cause accuracy vs 32% for a limit-check console; 70 s median detection.

**LineCheck · AI aircraft inspector on FaceTime**

Beta Fund × AWS × OpenAI hackathon · Oct 2026 · OpenAI Realtime, LiveKit

Technicians call it with hands on the aircraft: it inspects a photo of the part, cites FAA AC 43.13-1B, checks airworthiness directives and drafts the work order for a mechanic to sign.

**ClaimSaathi · health-insurance claim companion**

Paytm Build for India AI Hackathon, Bengaluru · Oct 2026 · rules engine, vision AI, voice

Reads the hospital bill, predicts the payout with a policy clause behind every cut, explains it in Hindi or English, and drafts the grievance when a settlement breaks IRDAI rules.

**Fitted · booking software from one paragraph**

Stripe × Supabase AI Payments Hack, Berlin · Oct 2026 · Stripe Connect, Supabase, OpenAI

One engine turns a business description into a padel club, spin studio or salon booking app with Stripe Connect payments; AI costs about $0.002 per call and never runs on a booking.

**Clinical ML · Sepsis, PharmaSentinel, Readmission**

Mar 2026 · XGBoost, Optuna, PyTorch, BERT, scikit-learn, SHAP · Amazon ML Summer Program

Sepsis ~0.91 AUROC seven hours before onset from 44 ICU signals; multi-task BERT across 215K+ drug reviews; 30-day readmission risk across 130 US hospitals.

### PROGRAMS

Google Cloud Gen AI Academy APAC, Code Vipassana and Startup School: multi-agent systems with ADK, Gemini, Vertex AI, AlloyDB and BigQuery · McKinsey Forward Program

### HOW EVERY PROJECT WORKS

| 01 · SENSE | 02 · MODEL | 03 · VERIFY | 04 · DECIDE |
|---|---|---|---|
| **Read the signal** | **Find the pattern** | **Prove it can be trusted** | **A human signs** |
| telemetry · vitals · images | XGBoost · LSTM · BERT | root causes · SHAP · citations | AI drafts, people decide |

**Scan for the full portfolio**

harshitaadlakha.github.io
Research, eight projects, live demos and replays.

### FIVE HACKATHONS, ONE WEEK

22 Oct · Paris · FDIR Copilot
23 Oct · San Francisco · LineCheck
24 Oct · Berlin · Fitted · London · Canva Gen©
27 Oct · Bengaluru · ClaimSaathi (Paytm)